How the Azure UAT database and web app were created

Veld Supply shop. Done on 28 October 2026 in the Azure portal, on the free account. These are the clicks that were used. Passwords and the JWT signing key are not written here.

1. Azure account

1. Open the Azure account page: `https://azure.microsoft.com/en-gb/pricing/purchase-options/azure-account/`
2. Choose **Try Azure for free** (the Azure free account column). Do not choose **Pay as you go**. Pay as you go has no spending cap.
3. Finish signup. Microsoft may place a temporary USD 1 check on the card and then reverse it. The free account includes USD 200 of credit for 30 days. That credit was not used for these two resources. Both were created on free plans.

2. SQL database

In the portal search box, open **SQL databases** and choose **Create**.

Basics

1. **Resource group:** create a new group named `veld-supply`. The web app uses this same group.
2. **Database name:** `EcommUat`.
3. **Server:** create a new server named `e-comm-uat`.
4. **Authentication: SQL authentication**. The API connects with a username and password. Microsoft Entra-only authentication would not work with the current connection string. "Use both SQL and Microsoft Entra authentication" is also acceptable.
5. **Server admin login:** `ecommuat`. The password was chosen in the portal and stored outside this document.
6. **Location: South Africa North**. This was the region the free offer allowed, and the web app was put in the same region.

Configure database

On **Compute + storage**, open **Configure database**.

1. Turn the **free database offer** on.
2. Set **Behavior when free limit is reached** to **Auto-pause the database until next month**. The other choice starts charging, and it cannot be switched back.
3. Leave the service tier on **General Purpose, serverless**. Do not switch it to a DTU or provisioned plan.
4. The cost summary on the right must say **0**. If it shows a monthly price, cancel and do not create the database.
5. Choose **Apply**.

The free offer is 100,000 vCore-seconds and 32 GB of data per month. The database also pauses after a period with no connections. The first request after that waits while it wakes up.

Networking

1. **Connectivity method:** Public endpoint. The free web app cannot use a private endpoint.
2. **Allow Azure services and resources to access this server:** Yes. That is how the site reaches the database.
3. **Add current client IP address:** Yes. That allows this PC to connect later. It opens that one address, not the whole internet.
4. **Minimum TLS:** 1.2.
5. **Connection policy:** Default.
6. Leave private access empty. Do not add a firewall rule from `0.0.0.0` to `255.255.255.255`.

The template's rule from `0.0.0.0` to `0.0.0.0` is the "allow Azure services" switch. It does not open the database to the internet.

Security

1. **Microsoft Defender for SQL:** Off. The trial turns into a paid plan.
2. **Transparent data encryption:** On, with the service-managed key.
3. **Ledger, auditing, and a customer-managed key:** Off.

Review and create

1. On **Review + create**, confirm the estimated cost is **0**, the free offer is applied, and the free-limit behavior is **Auto-pause**.
2. Choose **Create** and wait until the deployment shows **Succeeded**.

What was created

Setting	Value
Resource group	veld-supply
Server	e-comm-uat.database.windows.net
Database	EcommUat
Region	South Africa North
Tier	General Purpose, serverless (GP_S_Gen5)
Free offer	On, auto-pause when the monthly limit is reached
Max size	32 GB
Backup	Locally redundant
Public network	Enabled
TLS	1.2
Defender	Off

3. Web app

In the portal search box, open **App Services**, then **Create**, then **Web App**.

1. **Resource group:** `veld-supply`, the same group as the database.
2. **Name:** `veld-supply`.
3. **Publish:** Code.
4. **Runtime stack:** .NET 10. Preview in the list is fine.
5. **Operating system:** Windows.
6. **Region:** South Africa North, the same region as the database.
7. **Pricing plan: Free F1**. The default is often a paid Basic plan. Do not create the app unless F1 is selected.
8. If the wizard asks to create a database, choose **None**. `EcommUat` already exists. The wizard would create a second database that is not on the free offer.
9. On the deployment step, leave **continuous deployment** off. The repository already has a GitHub workflow. The portal would add a second one.
0. Skip Application Insights.
1. Choose **Create** and wait until the deployment shows **Succeeded**.

What was created

Setting	Value
App name	veld-supply
Resource group	veld-supply
Plan	Free F1 (ASP-veldsupply-a5c7)
Runtime	.NET 10
Region	South Africa North
Always on	Off (not available on the free plan)
Default domain	veld-supply-cpdddtaqf8epdpe0.southafricanorth-01.azurewebsites.net

The short name `veld-supply.azurewebsites.net` does not resolve. Open the site from the address above.

Where to open it

1. Go to `https://portal.azure.com`.
2. Search for `veld-supply`.
3. Open the result whose type is **App Service**, not the SQL server and not the resource group.
4. On **Overview**, click **Default domain**.

Until the shop is published, that address shows Azure's empty welcome page.

4. Settings added after creation

On the App Service, open **Settings**, then **Environment variables**. Leave **Deployment slot setting** unchecked. The free plan has no deployment slots.

Connection string

Add one connection string. It does not go in App settings.

- **Name:** `Shop`
- **Type:** SQLAzure
- **Value:** `Server=tcp:e-comm-uat.database.windows.net,1433;Initial Catalog=EcommUat;User ID=ecommuat;Password=YOUR_PASSWORD;Encrypt=True;TrustServerCertificate=False;Connection Timeout=30;`

`YOUR_PASSWORD` is the SQL admin password from the server setup. It is not stored in git.

App settings

These three go under **App settings**, not under connection strings.

- `Jwt__Issuer` = `veld-supply`
- `Jwt__Audience` = `veld-supply`
- `Jwt__Key` = a new random string, at least 32 characters, different from the local development key

Save the settings. The local development key stays in the development settings file. The Azure key stays only in Azure.

5. Publish profile

Download publish profile on Overview fails while basic authentication is off. New App Service apps start with it off.

1. Open **Settings**, then **Configuration**, then **General settings**.
2. Set **SCM Basic Auth Publishing Credentials** to **On**. Leave FTP off.
3. Save. Restart the app if the portal asks.
4. Return to **Overview** and choose **Download publish profile**. The file is `veld-supply.PublishSettings`.

That file is a password for deploying the site. It was saved as the GitHub Actions secret `AZURE_WEBAPP_PUBLISH_PROFILE` on the repository `JumariBlignaut/e-comm`. It was not committed.

> A push to the `release` branch is what publishes the shop to this web app. Creating the database and the web app does not deploy the code by itself.